INCH-POUND

MIL-C-24640/9C
26 June 1995
SUPERSEDING
MIL-C-24640/9B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI PAIR, AWG 22, TYPE 2XS (INCLUDING VARIATION TYPE 2XSA AND 2XSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size 22, maximum diameter 0.033 inches

SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Two conductors, one black and one white, twisted together. The pitch shall be 1½" ± ¼" (inch).

FOURTH - Binder tape applied helically with overlap, at manufacturer's option.

FIFTH - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

SIXTH - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.

AMSC N/A FSC 6145

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SEVENTH - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

EIGHTH - Binder tape applied helically with overlap.

CONSTRUCTION VARIANT FOR TYPE 2XS

NINTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE 2XSA

NINTH - Crosslinked polyolefin jacket. (See Table II for thickness).

TENTH - Braided metal armor, dimensions in accordance with Table II.

CONSTRUCTION VARIANT FOR TYPE 2XSO

NINTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

TENTH - Binder tape applied helically with overlap, at manufacturer's option.

ELEVENTH - Crosslinked polyolefin jacket. (See Table III for thickness).

TABLE I. Construction details - unarmored.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UN	2XS-2	2	0.038	0.308	0.332	61.4
-02UN	2XS-3	3	.038	.325	.350	74.2
-03UN	2XS-7	7	.041	.423	.455	135.0
-04UN	2XS-10	10	.044	.537	.579	194.0
-05UN	2XS-14	14	.044	.582	.627	245.0
-06UN	2XS-19	19	.044	.644	.694	319.0
-07UN	2XS-24	24	.047	.758	.818	431.0
-08UN	2XS-30	30	.047	.804	.866	500.0

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TABLE II. Construction details - armored.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01AN	2XSA-2	2	0.038	0.358	0.382	81.4
-02AN	2XSA-3	3	.038	.375	.400	101.7
-03AN	2XSA-7	7	.041	.473	.505	176.1
-04AN	2XSA-10	10	.044	.587	.629	228.5
-05AN	2XSA-14	14	.044	.632	.677	280.0
-06AN	2XSA-19	19	.044	.694	.744	359.5
-07AN	2XSA-24	24	.047	.808	.868	450.8
-08AN	2XSA-30	30	.047	.854	.916	531.3

Table III. Construction details - overall shield.

PIN M24640/9	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSO-3	3	0.271	0.038	0.380	0.410	117.0
-02UO	2XSO-7	7	.373	.041	.474	.510	190.0
-03UO	2XSO-10	10	.487	.044	.594	.640	287.0
-04UO	2XSO-14	14	.541	.044	.636	.686	323.0
-05UO	2XSO-19	19	.607	.044	.709	.765	423.0
-06UO	2XSO-30	30	.777	.047	.869	.937	644.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.

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Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 Khz spark test 5.5 KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.37
Voltage withstand - volts, root mean square, minimum	
* Conductor to conductor and shield	1000
Conductor to shield	Not applicable
* Shield to shield (component shield only)	200
* Component shield to overall shield (type 2XSO only)	500
Insulation resistance - megohms/1000 feet, minimum	
* Conductor to conductor and shield	2500
* Shield to shield (component shield only)	100
* Shield to shield (component shield to overall shield 2XSO only)	100
Conductor and shield continuity	No failure
Jacket flaws (2XSO only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	50
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	40 min, 63 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure

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	Group B:	
	Cold bending	No failure
*	Cross link proof test (percent maximum)	
	Jacket	50
*	Drip	Not applicable
	Tear - pounds per inch thickness, minimum	35
	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	3
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged, minimum	50
*	Permanence of printing (conductor) - cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - (type 2XSA only) - conformance to material, construction and coverage	No failure
	Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (type 2XSO only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D:	
	Flame Propagation	No Failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 2 - 14 500 feet nominal.
19 - 30 250 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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